

Algorithm 6

The Complete Internal Search Pipeline

Egyptian medicine-name search engineering review

5 August 2026

From raw text to ranked medicine families, confirmation, and abstention

What Algorithm 6 Receives and Returns

Input

A raw query string or structured unreadable-text request.

```
MYMLAX or text: RIV
unreadable_mode: after
```

The hidden benchmark target is never passed to search.

Catalog

25,066 product records are grouped into **17,476 commercial-name families**. Ranking happens at family level; package variants remain attached for display.

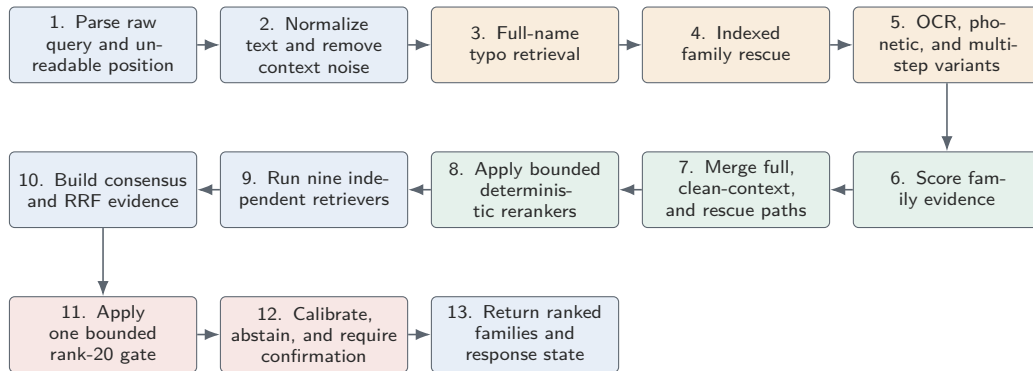
Output

- ▶ up to 20 ranked medicine families;
- ▶ variants, ingredients, examples, and evidence reasons;
- ▶ ambiguous or no_match response status;
- ▶ optional likely-match label that still requires confirmation.

Safety contract

The system retrieves candidates. It never certifies a prescription or permits automatic dispensing. Every positive result still requires user confirmation.

The Entire Algorithm, Flattened



The next slides open every numbered box. No predecessor algorithm is treated as an unexplained component.

Offline Catalog Preparation, With Stored Examples

Preparation runs once. The examples below are the actual forms reused at query time, not descriptions of a hidden model.

Representation	Stored example	Lookup created	Query it can support
Normalized name	CLEXANE → CLEXANE	exact and token maps	clexane finds the same family
Compact name	CLEX-A NE → CLEXANE	punctuation-free exact key	spacing/OCR artifact CLEX A NE
Prefix and suffix Character pieces	RIVOTRIL, reversed LIRTOVIR MYOLAX → MY, YO, OL, LA, AX	prefix and reversed-prefix maps 2-, 3-, 4-gram and BM25+ matrices	RIV or visible ending TRIL MYMLAX still shares MY, LA, AX
Consonant shape	KETOROLAC → KTRLK	skeleton and length buckets	KITORILIC also becomes KTRLK
Pronunciation keys	PHARMA, FARMA → PRM	medicine key, Match Rating, NYSIIS, Soundex	PH/F sound substitution
Deletion forms	delete one character from each family key	delete-key map and SymSpell radius 3	LANYU can reach LANTUS candidates
Family structure	ABIMOL EXTRA with head ABIMOL	family head, variants, ingredients, examples	visible head ABIMOL retrieves its longer family

Size after grouping 25,066 product rows become 17,476 commercial-name families; package variants stay attached to the family output.

Parse the Query and the Unreadable Position

The parser accepts ordinary text and explicit information about unseen characters. Position changes which catalog relationship counts as evidence.

Mode	Input example	Required catalog shape	Interpretation
none	pndol	ordinary fuzzy match	all visible characters form the query
after	ABIMOL [?]	candidate starts with ABIMOL and is longer	unreadable characters follow
before	[?] MOL	candidate ends with MOL and is longer	unreadable characters precede
middle	AB [?] OL	candidate starts with AB, ends with OL, and is longer	two visible anchors surround an unreadable region

Why this is not a wildcard everywhere

The location is explicit. A prefix claim cannot silently become a suffix or middle claim, which would multiply unrelated candidates.

Normalization, One Transformation at a Time

Normalization creates two representations because token scorers need spaces while character scorers do not.

Step	Input	Output	Reason
1	Clex-A NE 40 mg tab	CLEX-A NE 40 MG TAB	uppercase makes case irrelevant
2	CLEX-A NE 40 MG TAB	CLEX A NE 40 MG TAB	punctuation becomes one space
3	repeated or edge spaces	one internal space, no edge space	token positions become stable
4a	normalized text	CLEX A NE 40 MG TAB	preserve tokens for WRatio and token-sort
4b	normalized text	CLEXANE40MGTAB	remove separators for edit, n-gram, and phonetic search

Second example mymlax → normalized MYMLAX → compact MYMLAX; no character is corrected during normalization.

Boundary

Normalization changes representation only. It does not know that MYMLAX should retrieve MYOLAX; retrieval handles that later.

Context Cleanup, Token by Token

The original query is always searched. A second query is created only when recognized dosage, unit, route, or form tokens are present.

Normalized token	Class	Action	Running cleaned output
CLEX, A, NE	possible name text	keep	CLEX A NE
40	number next to a unit	remove	CLEX A NE
MG	recognized unit	remove	CLEX A NE
TAB	recognized form	remove	CLEX A NE

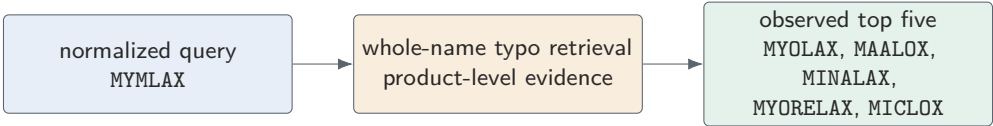
Example A CLEX-A NE 40 MG TAB → CLEX A NE.

Example B AUGMENTIN 1 GM TAB → AUGMENTIN.

Both raw and cleaned forms are searched. If no recognized context exists, as in MYMLAX, no second pass runs. A leading numeric brand token is kept unless surrounding context proves it is a dose.

Full-Name Typo Retrieval

The first path sends the normalized full query to the fast product-name retriever. It returns product candidates before family rescue starts.



Input	Full-name result	What succeeded	Why another path may still run
MYMLAX	MYOLAX rank 1	one middle replacement	family evidence still audits the order
CLEX-A NE 40 MG	CLEXANE rank 1 after	context removal exposes	raw and cleaned lists are both retained
TAB	cleaned pass	the brand	
LANS	target family may be	four visible characters only	short-head and delete-key rescue add candidates
	weak		

Indexed Family Rescue, Lookup by Lookup

Rescue runs when the full-name list is weak or uncertain, after context cleanup, for unreadable-position requests, and for three- or four-character queries.

Lookup	Query example	Catalog evidence	Candidate produced
Exact compact	CLEX A NE	compact key CLEXANE	family CLEXANE
Prefix	RIV	families beginning RIV . . .	RIVO, RIVOTRIL, others
Reversed suffix	MOL	reversed key begins LOM	ABIMOL families
Character grams	MYMLAX	shared MY, LA, AX	MYOLAX enters the pool
Consonant skeleton	KITORILIC→KTRLK	KETOROLAC→KTRLK	wrong-vowel candidate
Phonetic bucket	FARMA→PRM	PHARMA→PRM	sound-preserving candidate
Delete key	LANYU deletion forms	overlap with LANTUS deletion forms	LANTUS candidate
Family head	ABIMOL	validated head of ABIMOL EXTRA	longer family with variants

No lookup determines rank. Each one contributes family IDs to a union that is scored in later stages.

Bound the Rescue Candidate Pool Before Scoring

Indexed retrieval can return thousands of families for a short query. The prefilter removes families with no plausible visible evidence.

Step	Input example	Test	Output example
1	union for LANS	first character equal or confusable	discard unrelated initial letters
2	remaining family IDs	length within the configured local window	retain LANTUS-length names
3	survivors	shared grams, skeleton, phonetic, or head evidence	keep families with an explainable path
4	still too many IDs	stable family-ID order	cap at 2,200 before expensive scoring

Why the cap is safe but imperfect The cap controls runtime. Removing compatible-length scanning still lowers OCR Hit@20 from 73.28% to 71.77%, so prefilter recall remains measurable.

Generate General OCR and Typo Variants

Variants are generated from reusable error mechanisms. They are never keyed to a medicine name.

Mechanism	Reusable rule	Input → generated form	Candidate evidence
Keyboard/order	adjacent key, repeat, omission, adjacent swap	TAVANCI→TAVANIC	transposition candidate
Visual digit	0/O, 1/I/L, 2/Z, 3/E, 4/A, 5/S, 6/G, 8/B	COLCHICINE→COLCHICINE	exact transformed family key
Ligature	RN/M, CL/D, IV/N in both directions	RNI...→MI...	merged-stroke candidate
Phonetic	PH/F, CK/K, X/KS, QU/KW, GH/G, SH/CH	COUFSED→COUGHSED	pronunciation-preserving candidate
Vowel frame	replace or omit vowels, retain consonant order	KITORILIC→KETOROLAC	both skeletons are KTRLK
Multi-step	bounded combinations of two or three rules	uraranx→ORAMAX	synthetic non-regression chain

Guard against combinatorial explosion

Every generator has a fixed output limit and an evidence discount. A generated form adds candidates; it does not overwrite the query or automatically become rank 1. No table contains medicine-specific query exceptions.

Family Score: Exact Formula and One Full Calculation

For query q and candidate family f , every evidence term is normalized to $[0, 1]$ before applying its fixed coefficient.

$$\begin{aligned} S(q, f) = & 1.25E + 0.58L + 0.52L_w + 0.16P + 0.10S_f \\ & + 0.10G_2 + 0.18G_3 + 0.16C + 0.14H \\ & + 0.10U + 0.24A + 0.16V. \end{aligned}$$

Pair	E	L	L_w	P	S_f	G_2	G_3	C	H	U	A	V
MYMLAX/MYOLAX	0	.833	.833	.333	.500	.429	.143	1	1	0	.833	1

The weighted sum is about 1.75. Same first character adds 0.12, and equal length with $L_w \geq 0.76$ adds 0.18, producing about 2.05. It exceeds the 0.62 admission threshold. The score admits MYOLAX; later reranking decides its position.

Family Score Features, Edit and Boundary Evidence

Each row shows the exact input to one feature and the value it contributes before weighting.

Term	Input pair	Calculation	Output and meaning
E	CLEXANE/CLEXANE	compact strings equal	1.000, exact family
L	MYMLAX/MYOLAX	$1 - 1/6$	0.833, one ordinary edit
L_w	KITORILIC/KETOROLAC	lower costs for learned vowel confusions	0.794 versus raw 0.667
P	RIV/RIVOTRIL	3 shared leading characters divided by 8	0.375, visible prefix
S_f	MOL/ABIMOL	3 shared ending characters divided by 6	0.500, visible suffix
G_2	MYMLAX/MYOLAX	shared {MY,LA,AX}; union size 7	$3/7 = 0.429$
G_3	same pair	only LAX shared; union size 7	$1/7 = 0.143$

Family Score Features, Shape and Sequence Evidence

Term	Input pair	Transformation or calculation	Output and meaning
<i>C</i>	KITORILIC/KETOROLAC	remove vowels, merge confusion groups	both become KTRLK; $C = 1$
<i>H</i>	PHARMA/FARMA	PH/F and sound groups	both become PRM; $H = 1$
<i>U</i>	RIV/RIVOTRIL	letters appear in order over a span of 3	density 1, coverage 3/8, so $U = 0.75$
<i>A</i>	MYMLAX/MYOLAX	5 equal positions divided by maximum length 6	$A = 0.833$
<i>V</i>	MYMLAX/MYOLAX	shorter length divided by longer length	$V = 6/6 = 1$
<i>V</i>	RIV/RIVOTRIL	3/8	0.375, warns that most text is unseen

Why features are combined RIV has perfect visible order but only 37.5% length coverage. Sequence evidence alone must not certify RIVOTRIL.

Score Bonuses, Penalties, and Admission Examples

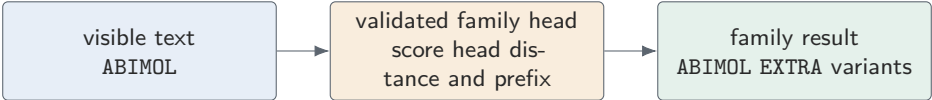
The weighted sum is adjusted only when a defined evidence pattern occurs.

Rule	Concrete pair	Adjustment	Why
same first character	MYMLAX/MYOLAX	+0.12	both start M
confusable first character	K.../C...	+0.06	C/K/Q share a configured confusion group
near-equal length + strong weighted edit	MYMLAX/MYOLAX	+0.18	equal length and $L_w = .833$
strong prefix + edit	a long candidate with most of its prefix visible	+0.08	two signals agree
strong edit + position	one mild OCR substitution	+0.22	$L_w \geq .84$ and $A \geq .70$
weak partial prefix	RIV/RIVOTRIL	at least -0.18	candidate is much longer than visible text
short query + weak edit	H/RIVO	-0.22	one character cannot identify a family
weak edit and weak sound/shape	unrelated family	-0.20	no independent support
weak prefix, grams, and edit	unrelated family	-0.10	all lexical channels are weak

Keep score ≥ 0.62 . A positional OCR exception can pass at 0.58 only when the first character, positional score, and visible-length coverage also pass.

Score Family Heads Without Collapsing Unrelated Drugs

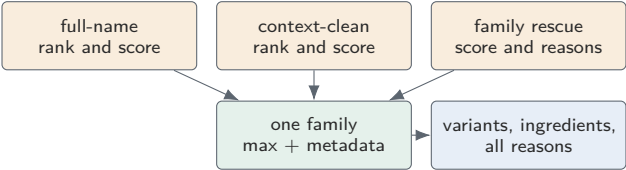
A package can contain a stable brand head plus a variant suffix. Head scoring lets visible brand text retrieve the family, but grouping requires validated catalog structure.



Visible query	Catalog structure	Action	Reason
ABIMOL	validated head of ABIMOL EXTRA	score the head and return the family	user may see only the stable brand head
LANS	short head evidence for LANTUS	admit as a candidate, not certainty	removal loses LANS→LANTUS from top 20
ABC	several unrelated ABC ... families	keep separate candidates	a shared token is not proof of one variant family

Merge Three Deterministic Evidence Paths

Candidates are merged by compact family key, not by product row.



Query	Evidence before merge	Merge operation	Final family record
MYMLAX	full path: MYOLAX rank 1; rescue: edit and n-gram evidence; no clean pass	group by compact key; keep maximum score and every reason	one MYOLAX record; agreement bonus +0.08
CLEX...40 MG TAB	full path: noisy list; clean path: CLEXANE rank 1; rescue: exact compact family	group by compact key; attach product variants to the family	one CLEXANE record; full/clean bonus +0.06

Deterministic Reranking Uses Bounded Corrections

The default order is score descending, clarification state, then stable name. Edit distance is never used as one global tie-break rule.

Correction class	Evidence required	Concrete example	Bound
Unique nearest	decisive raw or weighted distance advantage	MYOLANA→MYOLAX	equal-distance ties do not force a move
Exact visual/ligature	transformed key exactly reaches a family	COLCHICINE→COLCHICINE	distance and score limits must also pass
Phonetic + position	sound key and same-position characters agree	ECOFIXIM→CEFIXIME	phonetic key alone cannot promote
Skeleton + position	consonant frame and visible positions agree	KITORILIC→KETOROLAC	wrong-vowel evidence needs anchors
Family head	validated head plus compatible family	LANYU→LANTUS	unrelated prefixes stay separate
Multi-step exact chain	bounded chain reaches an exact family key	uraranx→ORAMAX	restore prior top if advantage is weak

Rank-1 protection

Fallback and generated candidates cannot change the first result without a unique, rule-specific advantage. This guard matters empirically: removing all deterministic reranking lowers OCR Hit@1 from 50.43% to 40.52%.

Independent Retrievers, Each With an Observed Example

After the deterministic family list is ready, nine independent lists search the same 17,476 families. Each returns at most 20 names.

Retriever	Input → representation	Observed output	What the vote means
Jaro–Winkler	compact TEFA against compact families	TELFAS source rank 1	order and common prefix agree
Match Rating	ABASCOLOGY → compressed pronunciation code	ABASAGLAR rank 1	consonant pronunciation agrees
RapidFuzz WRatio	normalized LACTULOSE; direct and partial ratios	LACTO rank 3	candidate is a strong partial form
NYSIIS	MYOLOGY → pronunciation-oriented code	MYOLAX rank 1	an independent sound encoding agrees
BM25+ 3-grams	LACTULOSE → overlapping three-character pieces	LACTO rank 12	informative local pieces survive

Remaining Independent Retrievers, With Examples

Retriever	Input $\rightarrow$ representation	Observed output	What the vote means
SymSpell	compact OSFOCU; symmetric deletes to distance 3	OSTOCAL rank 1	bounded edit candidate exists token/string view supplies a second vote broad phonetic backstop agrees shared TE and FA supply a third vote
RapidFuzz token-sort	normalize and sort tokens before ratio	TELFAS rank 2 for TEFA	
Soundex	TELBACE $\rightarrow$ coarse sound code	TELFAS rank 1	
Dice 2-grams	TEFA $\rightarrow \{TE, EF, FA\}$	TELFAS rank 15	

The deterministic family list is source 10. These nine methods run in parallel; none filters the next one. One source can be wrong without deleting candidates from the other lists.

Build One Consensus Record per Family

All ten top-20 lists are merged by compact family key. A candidate stores every source rank and four query-to-family similarities.

Stored features

- ▶ source count C ;
- ▶ reciprocal-rank fusion score;
- ▶ Levenshtein similarity;
- ▶ Jaro–Winkler similarity;
- ▶ character-bigram Dice;
- ▶ learned OCR-confusion similarity, diagnostic only.

Reciprocal-rank fusion

$$\text{RRF}(f) = \sum_{s \in S_f} \frac{1}{60 + r_{s,f}}$$

$r_{s,f}$ is the rank in source s . Rank 1 contributes $1/61 = 0.01639$; rank 14 contributes $1/74 = 0.01351$. The constant 60 makes several independent votes matter more than one exceptional rank.

Real record	Sources	RRF	Lev.	Jaro	Dice
MYMLAX→MYOLAX	9 of 10	.1439	.8333	.9111	.600

The nine source ranks are 1, 14, 1, 1, 2, 1, 1, 3, and 1. Learned-confusion similarity is 0.86 and remains diagnostic in the current rank-20 policy.

Consensus Ordering Is Explicit and Deterministic

The candidate union is sorted lexicographically by this tuple:

$$(-C, -RRF, -L, -J, \text{ name, family key}).$$

1. More independent sources wins first.
2. Better rank positions across those sources win next.
3. Levenshtein and Jaro–Winkler resolve remaining evidence ties.
4. Name and family key make the output stable across runs.

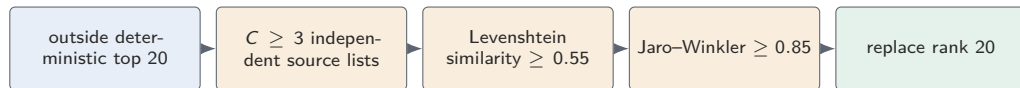
Query MYMLAX	sources	RRF	Lev.	Jaro	Order reason
MYOLAX	9	.143876	.833	.911	more sources
MYORELAX	7	.109917	.625	.856	remains below

What this order does not do

The consensus order does not replace the deterministic first result. A tested broad rank-1 consensus rule gained 27 synthetic cases and lost 138, so it remains disabled. Consensus is used only as a bounded retrieval rescue.

The Only Enabled Consensus Rewrite: One Rank-20 Slot

The deterministic order is copied first. At most one outside consensus candidate may replace the final slot.



Input → candidate	sources	Lev.	Jaro	Pass?	Result
TEFA→TELFAST	3	.571	.886	yes	inserted at rank 20
OSCAR→OSTOCAL	2	.571	.832	no	source and Jaro gates fail
NEUROTROPHIC→NEUROCET	4	.417	.850	no	Levenshtein gate fails
RIV→RIVOTRIL	2	.375	.854	no	source and Levenshtein fail

The first 19 families remain unchanged. This gate adds 2 OCR Hit@20 cases with 0 paired losses and changes no synthetic top- k result. The broader rank-1 gate is disabled in the policy file.

Worked Example: TEFA to TELFAST

This row shows why Algorithm 6 needs independent candidate sources.

Step	Observed execution
1. Parse and compact	TEFA stays TEFA; no context or unreadable marker.
2. Deterministic retrieval	TEFAST is outside the top 20; top 1 is TERA.
3. Independent lists	Jaro–Winkler rank 1, token-sort rank 2, Dice bigrams rank 15.
4. Source agreement	$C = 3$, exactly the minimum required.
5. Lexical gates	Levenshtein = $1 - 3/7 = 0.571$; Jaro–Winkler = 0.886. Both pass.
6. Bounded rewrite	TEFAST replaces rank 20. Ranks 1–19 remain unchanged.
7. Final output	top 1 TERA; TELFAST at rank 20; status ambiguous.

Result Hit@20 changes from failure to success without claiming that TELFAST is certainly the intended medicine.

Worked Example: MYMLAX to MYOLAX

This row succeeds before the external slot, then consensus measures how strong the agreement is.

Character evidence

- ▶ one replacement, $M \rightarrow O$;
- ▶ Levenshtein similarity = $5/6 = 0.833$;
- ▶ Jaro–Winkler = 0.911;
- ▶ same length and strong positional agreement.

Source evidence

MYOLAX occurs in 9 of 10 lists: deterministic, BM25+, Dice, Jaro, Match Rating, NYSIIS, token-sort, WRatio, and SymSpell.

Final output

MYOLAX remains rank 1. No external slot is needed. Current policy returns `ambiguous`, `needs_clarification=true`, and probability 0.0 because the optional abstention model is not enabled.

Final Safety State and the Disabled Optional Calibrator

The code can calculate a calibrated correctness probability when the policy contains an abstention block. The current policy contains no such block, so calibration returns probability 0 and never changes the message.

Calibration inputs	Exact contents
Top-family evidence	source count, RRF, Levenshtein, Jaro–Winkler, bigram Dice
Top-1 versus top-2 gaps	source-count, RRF, Levenshtein, and Jaro differences
Response context	candidate count, query length, rank-1 promotion flag, external-slot flag
Optional model	standardized logistic regression over these values; never changes ranking
Current policy	no abstention configuration; probability 0.0 and likely-match flag false

Candidates exist

Status ambiguous; message asks the user to compare evidence and confirm the medicine. Every result keeps `needs_clarification=true`.

An experimental 0.63 calibrator was measured separately on provisional audit-clean subsets, but it is not present in the active policy and is not a clinical guarantee.

No candidate survives

Status `no_match`; example `H→` no result. One visible character is not converted into a forced medicine family.

How Every Score in the Next Slides Was Produced

No value below is copied from a paper's dataset. Every method was rerun locally against the same two frozen evaluation sets.

Evaluation set	Cases	One case means	Metric example
OCR primary fair unique	464	one distinct scored compact query-target pair; real-drug collisions excluded	MYMLAX→MYOLAX; rank 1 gives H@1=1
Synthetic clean core	66,257	one fair generated corruption-target pair after collision filtering	generated typo with its catalog family target

Exact score source

benchmark_04_experiments/results/07_algorithm_6/ all_evaluated_method_scores.csv. The method descriptions and URLs come from results/06_competitor_benchmark/competitor_inventory.csv.

Interpretation The GitHub or paper project supplies method provenance. The percentages are our Egyptian medicine benchmark results, not claims copied from those projects.

Algorithms 1 to 6 on the Same Frozen Cases

Method	OCR 464			Synthetic 66,257			OCR ms
	H@1	H@5	H@20	H@1	H@5	H@20	
Algorithm 1	23.71	31.47	38.15	83.86	89.72	92.23	1.28
Algorithm 2	25.86	43.75	51.29	92.41	96.33	96.88	4.75
Algorithm 3	25.65	40.95	53.02	91.84	96.18	97.12	6.84
Algorithm 4	47.63	61.85	71.12	97.19	99.24	99.52	14.89
Algorithm 5	50.43	63.58	72.84	98.19	99.65	99.9985	28.54
Algorithm 6	50.43	63.58	73.28	98.19	99.65	99.9985	48.50

Code source	Local implementation
A1, A2, A3	app/app.js; external_algorithms/english_search_algorithm_fast.py; master_commercial_name_search.py
A4, A5, A6	algorithm_4_commercial_name_search.py; algorithm_5_commercial_name_search.py; algorithm_6_consensus_search.py

Algorithm 6 adds two OCR top-20 recoveries over Algorithm 5 and changes no rank-1 result. Its additional retrievers increase mean OCR latency by 19.96 ms.

External Lexical Baselines, Locally Rerun

Method	OCR 464		Synthetic 66,257		Method source
	H@1	H@20	H@1	H@20	
SymSpell, distance 3	38.58	55.82	94.22	98.80	SymSpell
Damerau–Levenshtein	34.27	59.27	94.78	99.64	RapidFuzz
Exhaustive Levenshtein	34.27	59.27	94.05	99.44	RapidFuzz
Jaro–Winkler	32.11	67.46	89.57	98.43	Jellyfish
BM25+ character 3-grams	20.26	47.41	78.19	92.17	rank_bm25
Character 3-gram TF–IDF	18.32	47.41	75.21	91.66	scispaCy reference
RapidFuzz weighted ratio	11.21	56.25	55.73	97.20	RapidFuzz
Exact or prefix	2.59	3.02	3.00	3.01	local baseline

Observed pattern Jaro–Winkler has the strongest external OCR H@20 at 67.46%, while Damerau–Levenshtein is stronger on synthetic H@20. Algorithm 6 exceeds both on OCR because it combines family, character, sound, and safety evidence.

Biomedical, Hybrid, and Phonetic Baselines

Method	OCR 464		Synthetic 66,257		Method source
	H@1	H@20	H@1	H@20	
BioSyn-style, sparse weight .50	20.47	53.02	78.17	93.78	BioSyn
xMEN-style SapBERT + TF-IDF RRF	16.81	52.37	48.85	92.30	xMEN
SapBERT dense HNSW	11.42	38.58	27.67	56.45	SapBERT
Egyptian medicine phonetic baseline	6.90	23.28	69.66	92.07	local implementation

Why these scores are included

Dense biomedical normalization methods test semantic representation; sparse hybrids add character evidence; the phonetic baseline tests sound alone. Their low OCR H@1 shows that prescription corruption cannot be solved by semantic or phonetic similarity without family-aware lexical evidence.

External Method Provenance

Methods used locally	Original project or implementation reference
Levenshtein, Damerau, WRatio	github.com/rapidfuzz/RapidFuzz
Jaro–Winkler and phonetic encoders	github.com/jamesturk/jellyfish
SymSpell	github.com/wolfgarbe/SymSpell
BM25 family	github.com/dorianbrown/rank_bm25
BioSyn-style hybrid	github.com/dmis-lab/BioSyn
SapBERT dense retrieval	github.com/cambridgeltl/sapbert
xMEN-style dense+sparse RRF	github.com/hpi-dhc/xmen
Character TF–IDF reference	github.com/allenai/scispacy
Phonetic method reference	github.com/microsoft/PhoneticMatching

All adapters, catalog preparation, ranking, and scoring were executed by `run_competitor_benchmark.py`. Source links identify the method family; they do not imply that upstream authors evaluated this Egyptian catalog.

Ablation Contract and Top-Level Paths

Reference is complete Algorithm 6 on the same 464 OCR pairs: H@1 50.431%, H@20 73.276%. Each row removes only the named component, reruns the changed pipeline, then leaves all other retrieval and consensus stages active.

Removed component	H@1	Δ H@1	H@20	Δ H@20	Observed example
Nothing	50.431	0	73.276	0	MYMLAX→MYOLAX rank 1
Family rescue	36.638	−13.793	56.034	−17.241	TUTORIALAE→KETOROLAC: 7 to > 20
External full-name retriever	49.784	−0.647	72.414	−0.862	ABDUSONG LAR→ABASAGLAR: 1 to > 20
Context-clean second pass	50.431	0	73.276	0	no top- <i>k</i> change on these 464 pairs; CLEX...40 MG TAB still explains the path

Largest dependency The family-rescue path is responsible for most measured recall. Ablation effects overlap and therefore cannot be added.

Retrieval Ablations With Measured OCR Losses

Retrieval removed	H@1	H@20	Δ H@20	Changed input $\rightarrow$ target
Short-edge retrieval	49.784	70.259	-3.017	RICOHIL $\rightarrow$ RIVOTRIL: 3 to > 20
Family-head rescue	48.060	70.905	-2.371	LANYU $\rightarrow$ LANTUS: 2 to > 20
Delete-key retrieval	48.276	71.121	-2.155	LANYU $\rightarrow$ LANTUS: 2 to > 20
Compatible-length scan	49.784	71.767	-1.509	INDULGENCE $\rightarrow$ INDERAL: 10 to > 20
Short visible-head rescue	50.431	73.060	-0.216	LANS $\rightarrow$ LANTUS: 20 to > 20
Short-frame rescue	50.431	73.276	0.000	one H@20 gain and one loss cancel; LAE $\rightarrow$ LACTO is lost

H@20 after removal is the complete rerun score, not a standalone component accuracy. Shared cases explain why family-head and delete-key effects overlap.

Retrieval Ablations With Zero Net OCR Top- k Change

Every row below produces H@1 50.431% and H@20 73.276% after removal on the 464 OCR pairs.

Retrieval removed	Example mechanism	What zero means
All multi-step chains	uraranx→ORAMAX	other OCR sources cover this sample; synthetic H@1 loses 343 and gains 3
Bounded-nearest fallback	unique nearest candidate	no OCR top- k pair changed
Evidence-query variants	alternate evidence form	no OCR top- k pair changed
First-character expansion	confusable C/K/Q initial	no OCR top- k pair changed
Keyboard rescue	adjacent key or repeat	no OCR top- k pair changed
Ligature rescue	RN/M, CL/D, IV/N	no OCR top- k pair changed
Short-query rescue	three- or four-character pool	no net change beyond overlapping paths
Transposition rescue	TAVANCI→TAVANIC	no OCR top- k pair changed
Visual-deletion chains	digit/letter plus omission	no OCR top- k pair changed
Visual-phonetic chains	visual plus sound rewrite	no OCR top- k pair changed

Zero is sample-specific redundancy, not proof that the mechanism is useless.

Scoring-Signal Ablations, Largest Effects

Signal removed	H@1	Δ H@1	H@20	Δ H@20	Observed loss example
Weighted edit similarity	49.569	-0.862	66.164	-7.112	TUTORIALAE→KETOROLAC: 7 to > 20
Raw edit similarity	49.353	-1.078	67.888	-5.388	same case loses uniform edit support
Length coverage	50.216	-0.216	70.259	-3.017	15 H@20 losses, 1 gain
Positional evidence	48.922	-1.509	71.336	-1.940	TEFAXATE→TELFAS: 16 to > 20
Prefix evidence	48.707	-1.724	71.336	-1.940	RIVITAM→RIVOTRIL: 17 to > 20
Character n-grams	48.060	-2.371	71.983	-1.293	local pieces around corruptions disappear

Weighted and raw edit signals affect many of the same rows. Their deltas are paired ablations, not independent contributions to add together.

Scoring-Signal Ablations, Smaller or Rank-Only Effects

Signal removed	H@1	Δ H@1	H@20	Δ H@20	Observed effect
Weighted confusion costs	48.707	-1.724	72.845	-0.431	KETOMAN→KATOMOR: 14 to > 20
Consonant skeleton	49.784	-0.647	73.060	-0.216	one top-20 loss
Subsequence evidence	50.000	-0.431	73.060	-0.216	two losses and one gain
Phonetic evidence	49.569	-0.862	73.276	0	ECOFIXIM→CEFIXIME: rank 1 to 2
Retriever agreement	50.216	-0.216	73.276	0	OSSETIA→OSSICA: rank 1 to 3
Suffix evidence	49.569	-0.862	73.276	0	RIVOFLURAL→RIVOTRIL: rank 1 to 2

Signals with zero H@20 delta can still protect rank 1. H@1 and H@20 answer different questions and must both be reported.

Reranker Ablations

Reranker removed	H@1	H@20	Δ H@1	Observed effect
All deterministic reranking	40.517	70.690	-9.914	LANYU→LANTUS: 2 to > 20
Candidate-pool/head rerankers	48.276	71.336	-2.155	10 H@1 and 9 H@20 losses
Character-evidence rerankers	48.922	73.276	-1.509	RIVOFN2→RIVOTRIL: rank 1 to 2
Strict full-name correction	49.569	73.276	-0.862	MYOLANA→MYOLAX: rank 1 to 2
Exact-key rerankers	50.431	73.276	0	no top- <i>k</i> change on OCR 464
Multi-step-chain rerankers	50.431	73.276	0	no top- <i>k</i> change on OCR 464
Nearest-candidate rerankers	50.431	73.276	0	no top- <i>k</i> change on OCR 464
Weighted-edge rerankers	50.431	73.276	0	no top- <i>k</i> change on OCR 464

The large all-reranking drop includes overlapping correction families. It does not imply that every zero-delta subgroup can be deleted without a fresh test.

Score-Tuning, Safety, and Atomic-Flag Ablations

Removed item	H@1	H@20	Interpretation
Exact-chain score augmentation	50.431	73.276	no OCR top- <i>k</i> pair changed
In-place chain score tuning	50.431	73.276	no OCR top- <i>k</i> pair changed
Safety clarification gate	50.431	73.276	retrieval is unchanged; top- <i>k</i> cannot measure response safety
Strict full-name evidence guard flag off	50.647	73.276	2 H@1 gains and 1 loss, so net score hides a regression
Strict full-name rerank flag off	49.569	73.276	4 H@1 losses
Other 52 atomic flags, one at a time	50.431	73.276	no OCR top- <i>k</i> change individually

The 54 atomic rows test implementation switches inside the semantic components. The complete auditable list remains in `algorithm_6_replay_metrics.csv`.

Algorithm 6 Consensus-Source Ablations

Removed item	H@1	H@20	$\Delta H@20$	Observed consequence
Bounded rank-20 slot	50.431	72.845	-0.431	TEFA and LACTULOSE leave top 20
Jaro-Winkler source	50.431	73.060	-0.216	TEFA→TELFAS loses one vote
RapidFuzz token-sort source	50.431	73.060	-0.216	same TEFA case loses one vote
Dice bigram source	50.431	73.060	-0.216	shared TE and FA vote disappears
BM25+ trigram source	50.431	73.060	-0.216	competing outsider replaces LACTO
Match Rating source	50.431	73.276	0	remaining sources preserve both accepted gains
WRatio source	50.431	73.276	0	remaining sources preserve both accepted gains
NYSIIS source	50.431	73.276	0	remaining sources preserve both accepted gains
SymSpell source	50.431	73.276	0	remaining sources preserve both accepted gains
Soundex source	50.431	73.276	0	remaining sources preserve both accepted gains

Single-source zero deltas show redundancy on this sample. They do not prove the source never contributes candidate evidence on unseen reviewed crops.

Failure Examples: Missing or Insufficient Evidence

The supplied family is outside all final top-20 lists in these rows.

Input	Supplied target	Top 1	Target evidence	What the failure means
H	RIVO	none	0 sources; one character	insufficient evidence; no-match is safer
HOAVACH	AMIKACIN	HYABAK	0 sources; distance 6	severe mixed corruption; retrieval, not reranking, fails
LANSOPRAZOLE	LANTUS	PANTOPRAZOLE	1 source; Lev. 33.3%; Jaro 81.1%	supplied target is much farther than visible alternatives
RIV	RIVOTRIL	RIVO	2 sources; Lev. 37.5%; Jaro 85.4%	exact short family and several prefixes compete
ABACAVIR	ABASAGLAR	TECAVIR	1 Soundex source; Lev. 31.6%	one phonetic vote cannot justify insertion

Aggregate Of 124 top-20 misses, 97 targets appear in none of the ten source lists. They cannot be repaired by changing the final sorter.

Failure Examples: Gate Rejection or Label Conflict

Input	Supplied target	Top 1	Target evidence	What the failure means
OSCAR	OSTOCAL	ASCOR	2 sources; Lev. 57.1%; Jaro 83.2%	Levenshtein passes; source count and Jaro fail
NEUROTROPHIC	NEUROCET	NEUROTOP	4 sources; Lev. 41.7%; Jaro 85.0%	only Levenshtein blocks four weak votes
LANLTA	LANTUS	LANETAL	2 sources; Lev. 33.3%; Jaro 80.6%	target is weak and returned family is one edit closer
TOCILIZUMAB	TOCO	ALIZOLAM	1 WRatio source; Lev. 27.3%; Jaro 67.4%	target discards most visible characters; audit the label
CINNARIZIN	MYOLAX	CINNARIZINE	0 sources; target distance 9; nearest 1	visible text strongly suggests a different catalog drug

Do not tune blindly Twenty-seven misses have some source support but fail a gate. Many also conflict with a closer catalog name, so crop review must precede threshold reduction.

Runtime and Query-Time Work

Part	Bounded work	Practical cost
Catalog preparation	build deterministic maps, sparse 2/3-gram matrices, phonetic arrays, and SymSpell dictionary	once per process for 17,476 families
Deterministic retrieval	indexed lookup plus bounded family scoring and correction rules	measured primary path mean: 28.54 ms/query
Independent retrieval	several catalog scans, two sparse matrix products, and SymSpell lookup	adds most Algorithm 6 overhead
Consensus merge	at most $10 \times 20 = 200$ source appearances; merge by family and sort	small relative to retrieval
Complete Algorithm 6	all stages, gates, and output formatting	measured OCR mean: 48.50 ms/query

Optimization target

Cache catalog preparation and parallelize independent source queries while requiring exact top-20 parity. Removing evidence merely to reduce latency can erase the two measured consensus gains.

Executable Algorithm Summary

1. Parse visible text and any before/middle/after unreadable marker.
2. Normalize the query; optionally create a context-clean second query.
3. Search the full catalog, then retrieve family candidates from exact, prefix, suffix, n-gram, skeleton, phonetic, delete, head, and length indexes.
4. Generate bounded keyboard, visual, ligature, phonetic, vowel, and multi-step variants.
5. Score candidates with raw and weighted edit, edges, n-grams, sound, consonants, subsequence, position, and length evidence.
6. Merge evidence by commercial family; apply bounded deterministic corrections and preserve weak cases as ambiguous.
7. Run Jaro–Winkler, Match Rating, WRatio, NYSIIS, BM25+ trigrams, SymSpell, token-sort, Soundex, and Dice bigrams.
8. Merge ten lists, compute source count and RRF, and allow one external rank-20 candidate only after the 3/0.55/0.85 gate.
9. Estimate likely correctness without changing ranking; require confirmation or abstain.
10. Return up to 20 family-level candidates with evidence and safety state.

The algorithm is a controlled evidence pipeline, not one distance formula and not one opaque earlier algorithm.

Algorithm 6, in one sentence

Generate medicine-family candidates from several independent representations, rank them with bounded character and structural evidence, add only one consensus-supported retrieval rescue, and never convert lexical agreement into unconfirmed clinical certainty.

Implementation: `benchmark_01_legacy/master_algorithms/algorithm_6_consensus_search.py`